Solution Requirements

SMARTBRIDGE - Solution Requirements

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	**Authentication** PDF	Users register using email and password. Passwords are encrypted using `bcryptjs`. Login issues a signed JWT token. DOCX+ 2	**High** PDF
2	**Authorization levels** PDF	Authenticated Users can manage their personal workouts and invoke AI features. Middleware restricts unauthorized API calls. DOCX+ 1	**High** PDF
3	**External interfaces** PDF	Integrates with **Google Gemini AI API** using the `@google/genai` SDK for workout generation and analytics. DOCX	**High** PDF
4	**Transactions processing** PDF	Handles transactional CRUD requests for workout records (create, read, search, update, delete) returning JSON responses. DOCX	**High** PDF

5	**Reporting** PDF	Generates performance reports including total workouts, average duration, and total calories burned via AI insights. DOCX	**Medium** PDF
6	**Business rules, etc.** PDF	Validation rules: Workout duration must be > 0, calories burned ≥0, and emails must be unique. DOCX+ 2	**High** PDF
7	**Compliance to laws** PDF	Ensures user privacy by salting/hashing passwords and avoiding exposure of credentials in API payloads. DOCX	**High** PDF
8	**Other** PDF	Full-text query searching for workouts based on workout name, category, or workout date. DOCX	**Medium** PDF

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category PDF	Requirement Description PDF	Target Metric / Acceptance Criteria PDF
1	**Performance & Speed** PDF	Fast response times for CRUD APIs and Gemini AI request processing. DOCX	API responses <200ms; AI generation <3 seconds. DOCX
2	**Scalability** PDF	Handles growing database records and concurrent user requests. DOCX	MongoDB Atlas cluster scales seamlessly up to 10,000 users. DOCX

3	**Security & Data Privacy** PDF	Secure transmission, encrypted stored passwords, and protected API endpoints. DOCX	Passwords hashed with bcrypt (salt 10); JWT authentication required. DOCX
4	**Reliability & Availability** PDF	High server uptime and resilient database connectivity. DOCX	99.9% uptime per month with automatic MongoDB Atlas connection retry. DOCX
5	**Usability & Accessibility** PDF	Standard RESTful API output structure and clear error responses. DOCX	100% standardized JSON format with status codes (200, 201, 400, 500). DOCX
6	**Maintainability**	Clean Code separation following the MVC architecture. DOCX	Modular separation across Controllers, Services, Models, and Routes. DOCX  DOCX FitSense - Personalized fitness recom